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DEPARTMENT:-

BS-CYBER SECURITY-FALL-24

SUBJECT:-

PROGRAMMING FUNDAMENTALS (Lab)

PRESENTED BY

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PROVIDED TO

Mr WASEEM YOUNAS

ASSIGNMENT #01

TASK – 1

Write a program that:

Prompts the user to enter two integers.
Performs and displays the results of the following operations:

- **Addition**
- **Subtraction**
- **Multiplication**
- **Division (handle division by zero)**

Code:

```
#include <iostream>
using namespace std;
int main ()
{
    int Add, Sub, Mul, Div, int1, int2 ;
    // Taking integers from user
    cout << "Enter two Integers " << endl ;
    cin >> int1 >> int2 ;

    // Performing Addition
    Add = int1 + int2 ;
    cout << "The Addition of Integers is " << Add << endl ;

    // Performing Subtraction
    Sub = int1 - int2 ;
    cout << "The Subtraction of Integers is " << Sub << endl ;

    // Performing Multiplication
    Mul = int1 * int2 ;
    cout << "The Multiplication of Integers is " << Mul << endl ;

    // Performing Division and handling with Zero

    if ( int2 == 0)
    {

        cout << "The Division is not Possible by Zero " << endl ;
    }
    else
    {
        Div = int1 / int2 ;
        cout  << "The Division of Intergers is " << Div << endl ;
    }
    return 0;
}
```

Testing with Different inputs

Output 1:

```
D:\University\Programming Fundamentals (Lab)\Assignments\Task 1.exe
Enter two Integers
96
8
The Addition of Integers is 104
The Subtraction of Integers is 88
The Multiplication of Integers is 768
The Division of Intergers is 12

--------------------------------
Process exited after 22.94 seconds with return value 0
Press any key to continue . . .
```

Output 2:

```
D:\University\Programming Fundamentals (Lab)\Assignments\Task 1.exe
Enter two Integers
12
0
The Addition of Integers is 12
The Subtraction of Integers is 12
The Multiplication of Integers is 0
The Division is not Possible by Zero

--------------------------------
Process exited after 5.111 seconds with return value 0
Press any key to continue . . .
```

TASK - 2

Develop a program that:
Asks the user for the length and width of a rectangle.
Calculates the area using the formula: *A=L x W*

Code:

```
Task 1.cpp    [*] Task 2.cpp
#include <iostream>
using namespace std;
int main ()
{
    //Declaring required variables
    float Length, Width, Area ;

    //Taking Length and Width from user
    cout << "Enter Length of the Rectangle " << endl ;
    cin >> Length ;

    cout << "Enter Width of the Rectange " << endl ;
    cin >> Width ;

    //Calculating the Area by given formula
    Area = Length * Width ;
    cout << "The Area of Rectange is " << Area << endl ;

    return 0 ;
}
```

Testing with different inputs

Output 1:

```
D:\University\Programming Fundamentals (Lab)\Assignments\Task 2.exe
Enter Length of the Rectangle
65
Enter Width of the Rectange
34
The Area of Rectange is 2210

--------------------------------
Process exited after 6.926 seconds with return value 0
Press any key to continue . . .
```

Output 2:

```
sk 1.cpp    Task 2.cpp
D:\University\Programming Fundamentals (Lab)\Assignments\Task 2.exe
Enter Length of the Rectangle
65.78
Enter Width of the Rectange
34.26
The Area of Rectange is 2253.62

--------------------------------
Process exited after 11.87 seconds with return value 0
Press any key to continue . . .
```

TASK - 3

Write a program that:
Converts a temperature from Celsius to Fahrenheit.
Asks the user to input the temperature in Celsius.
Use the formula: $F = (C * 9/5) + 32$

Code:

Task 1.cpp | Task 2.cpp | [*] Task 3.cpp

```cpp
#include <iostream>
using namespace std;
int main()
{
	//Declaring variables
	double celsius, fahrenheit ;
	
	//Taking temperature from user
	cout << "Enter Temperature in Celsius " << endl ;
	cin >> celsius ;
	
	//Calculating temperature in Fahrenheit
	fahrenheit = ( celsius *  9 / 5 ) + 32 ;
	cout << "The Temperature in Fahrenheit is "
		<< fahrenheit << endl ;
	
	return 0 ;
}
```

Testing with Different inputs

Output 1:

```
D:\University\Programming Fundamentals (Lab)\Assignments\Task 3.exe
Enter Temperature in Celsius
33
The Temperature in Fahrenheit is 91.4

--------------------------------
Process exited after 7.703 seconds with return value 0
Press any key to continue . . .
```

Output 2:

```
D:\University\Programming Fundamentals (Lab)\Assignments\Task 3.exe
Enter Temperature in Celsius
45.786
The Temperature in Fahrenheit is 114.415

--------------------------------
Process exited after 7.636 seconds with return value 0
Press any key to continue . . .
```

TASK – 4

Create a program that:

Asks the user for the principal amount, rate of interest, and time (in years).

Calculates the simple interest using the formula:

$Interest = (Pricipal * Rate * Time)\ /\ 100$

Code:

Task 1.cpp × Task 2.cpp × Task 3.cpp × Task 4.cpp ×

```cpp
# include <iostream>
using namespace std;
int main ()
{
    //Declaring variables
    double amount, rate, time, interest;

    //Taking required data from user
    cout << "Enter Principal Amount " << endl ;
    cin >> amount ;

    cout << "Enter Rate of Interest ( in % ) " << endl ;
    cin >> rate ;

    cout << "Enter Time in Years " << endl ;
    cin >> time ;

    //Calculating Simple Interest
    interest = ( amount * rate * time ) / 100 ;
    cout << "The Simple Interest is " << interest << endl ;

    return 0 ;
}
```

Testing with Different inputs

Output 1:

```
D:\University\Programming Fundamentals (Lab)\Assignments\Task 4.exe
Enter Principal Amount
30000
Enter Rate of Interest ( in % )
12
Enter Time in Years
2
The Simple Interest is 7200

--------------------------------
Process exited after 11.87 seconds with return value 0
Press any key to continue . . .
```

Output 2:

```
D:\University\Programming Fundamentals (Lab)\Assignments\Task 4.exe
Enter Principal Amount
60540
Enter Rate of Interest ( in % )
33
Enter Time in Years
4
The Simple Interest is 79912.8

--------------------------------
Process exited after 17.45 seconds with return value 0
Press any key to continue . . .
```

TASK – 5

Write a program that:

Asks the user to input the lengths of the three sides of a triangle.

Calculates the area using Heron's formula:

Calculate the semi-perimeter: $S=a+b+c2$

Use the semi-perimeter to calculate the area: $A=\sqrt{}S(S-a)(s-b)(s-c)$

Code:

Task 1.cpp × Task 2.cpp × Task 3.cpp × Task 4.cpp × Task 5.cpp ×

```cpp
#include <iostream>
#include <cmath>
using namespace std;
int main()
{
    //Declaring variables
    float side1, side2, side3, semi, area;

    //Taking length of each side from user
    cout << "Enter Length of First Side " << endl ;
    cin >> side1 ;

    cout << "Enter Length of Second Side " << endl ;
    cin >> side2 ;

    cout << "Enter Length of Third Side " << endl ;
    cin >> side3 ;

    //using Heron's Formula

    //Calculating the semi perimeter first
    semi = ( side1 + side2 + side3 ) / 2 ;
    cout << "The Semi-Perimeter is " << semi << endl ;

    //Using the semi perimeter to find the area
    area = sqrt ( semi * ( semi - side1 ) * ( semi - side2 ) * ( semi - side3 ) ) ;
    cout << endl << "The Area of Triangle is " << area ;

    return 0 ;
}
```

<u>Testing with Different inputs</u>

Output 1:

```
D:\University\Programming Fundamentals (Lab)\Assignments\Task 5.exe
Enter Length of First Side
3
Enter Length of Second Side
7
Enter Length of Third Side
5
The Semi-Perimeter is 7.5

The Area of Triangle is 6.49519
--------------------------------
Process exited after 3.89 seconds with return value 0
Press any key to continue . . .
```

Output 2:

```
D:\University\Programming Fundamentals (Lab)\Assignments\Task 5.exe
Enter Length of First Side
3.5
Enter Length of Second Side
5.9
Enter Length of Third Side
7.5
The Semi-Perimeter is 8.45

The Area of Triangle is 10.0661
--------------------------------
Process exited after 11.12 seconds with return value 0
Press any key to continue . . .
```